

Salon360: All-in-One Salon Management and Virtual Hair Try-On System

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Chapter 1

Introduction

This report outlines the approach we plan to take with our project and provides an overview of the targeted problem, the motivation behind it, the proposed solution, the scope of the project, the architectural view, and its use cases. This document is intended for the FYP committee, faculty members, developers, students, and researchers.

As virtual try-ons become increasingly popular within the fashion and beauty industry, with the rise of Generative AI, we focus on hair virtual try-ons that would satisfy clients and will empower them to make bold decisions.

Salon360 is a web application that aims to integrate a virtual hair try-on with a salon management system to create an all-in-one experience. This involves using advanced models and techniques to analyze the customer's uploaded photo to offer realistic virtual hair try-ons and give them personalized recommendations that provide a seamless and integrated experience.

1.1 Problem Statement

The idea of virtually trying on hairstyles or hair colors is not new, however, existing try-on systems fail to deliver personalised results. Most customers are unaware of their hair type and the ideal hairstyles that suit them. Moreover, current systems lack the option to try on advanced hair color treatments. Such systems are insufficient in a salon environment where customers expect realistic outputs.

Features	Garnier	Hairpaca	Beauti	Salon360
 Filter Layer: Feasibility check & show possible options	✗	✗	✗	✓
 Recommendation based on face shape	✗	✗	✗	✓
 Automated deal generation	✗	✗	✗	✓
 POS and management system	✗	✗	✓	✓
 Hair Style Try-on	✗	✓	✗	✓
 Reminders for bookings	✗	✗	✓	✓
 Hair Color Try-on	✓	✗	✗	✓

Figure 1.1: Comparison of features offered by our solution versus competitors: Garnier [2], Hairpaca [3], Beauti Software [1]

1.2 Scope

Salon360 is a web application that will combine virtual hair try-ons with a salon management system to create an efficient and unified experience for salons and their clients. It will include management tools that will handle salon operations such as booking appointments, and maintaining employee and customer records with POS integration.

The user module would provide three options to clients. The recommendation and try-on features will require clients to upload a photo of themselves and get recommendations based on face shape or try-on hairstyles and view realistic outputs. The virtual hair color try-on will range from basic to bold colors. The booking option is an integration into the management module.

To maintain the quality of our virtual try-on, the system will handle only front-facing 2D images with the hair and face completely visible. Any obstructions that cover the head like hats or cover the face like face masks will not be catered and the user would have to re-upload an image that follows the guidelines provided. The image must also be from an appropriate distance (similar to a passport photo but with the entire hair visible) and must be taken in ample light. The output would be given on the image the user uploaded, and no other angles would be provided. Additionally, our scope is limited to performing try-ons on static images only; real-time videos will not be supported.

1.3 Modules

This project is divided into two modules:

1.3.1 Module 1: Management Module (Admin Web App)

This module comprises a system to manage records of all salon operations such as customers, their bookings, salon services, and employee management. This will also include an AI driven automated deal generation module.

1. POS and Management System
 - Central database to maintain records and generate invoice
2. Automated Deal Generation
 - Automated promotions to meet financial targets and customer needs
3. Email Reminders for Bookings
4. Sync Try-On with Bookings

1.3.2 Module 2: User Module (Client Web App)

The user module gives a virtual experience for clients to visualize their looks before their bold transformations with an ideal recommendation system based on facial features.

1. Hair Color Try-On
 - Basic hair colors to advanced treatments (highlights and lowlights)
2. Hairstyle Try-On
 - Different hairstyle try-on catering to both genders
3. Recommendations Based on Face Shape

1.4 User Classes and Characteristics

This section identifies various salon360 user classes and their characteristics.

Table 1.1: Description of User Classes for Salon360

User class	Description
Salon Clients	Salon clients are regular salon goers that would be interested in virtually trying on hairstyles and hair colors before purchasing a service. They could also want to get a hairstyle recommendation. Clients are non-technical users who would prefer a convenient system with a user-friendly interface.
Salon Owners	Salon owners are the individuals with the power to integrate Salon360 in their salons. They would want a system that controls managerial operations such as booking appointments, employee information and transactions through a POS. Moreover, they would want a virtual try-on system that attracts customers and helps in ensuring their satisfaction. They would also appreciate automated deal generation to meet their sales targets.
Salon Employees	These include the receptionists that would want a management and POS system for a computerized and organized way to handle everyday operations. Customers will be able to book appointments themselves which would make the work of receptionists easier. Employees would also include stylists and other staff that would value a virtual try-on as it makes their task of explaining hairstyles or hair colors easier. Furthermore, the recommendation module would also help stylists assist customers in making decisions.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the Salon360.

2.1 Use-case and Event Response Table

2.1.1 High Level Use Case

Use Case (UC01)	Book Services
Actors	Clients (Users), System
Type	Primary
Description	The client selects a salon service for a walk-in appointment or for a future booking and the system records it in the database.

Table 2.1: High Level Use Case: Book Services

Use Case (UC02)	Virtual Hairstyle Try-On
Actors	Clients (Users), System
Type	Primary
Description	The client uploads a photo of themselves without any obstructions on their face and hair then chooses a hairstyle to try on. The system will apply the selected hairstyle on the image and give an output.

Table 2.2: High Level Use Case: Virtual Hairstyle Try-On

Use Case (UC03)	Overview Financial Records
Actors	Salon Owners, System
Type	Primary
Description	The salon owner accesses the sales records section of the system to review financial performance over a specified period (e.g., daily, weekly, monthly).

Table 2.3: High Level Use Case: Overview Financial Records

Use Case (UC04)	Get Hairstyle Recommendation
Actors	Clients (Users), System
Type	Primary
Description	The client uploads a photo of themselves without any obstructions on their face and hair. The system analyzes facial features and returns a hairstyle recommendation.

Table 2.4: High Level Use Case: Get Hairstyle Recommendation

Use Case (UC05)	Virtual Hair Color Try-On
Actors	Clients (Users), System
Type	Primary
Description	The client uploads a photo of themselves without any obstructions on their face and hair then chooses a hair color to try on. The system will apply the selected hair color on the image and give a realistic output.

Table 2.5: High Level Use Case: Virtual Hair Color Try-On

Use Case (UC06)	Generate Automated Deals
Actors	Salon Owners (Admins), System
Type	Secondary
Description	The salon owner can use the system to automatically generate deals to boost sales base on service sale trends.

Table 2.6: High Level Use Case: Generate Automated Deals

Use Case (UC07)	Manage Service Records (CRUD Operations)
Actors	Admin (Salon Owner), System
Type	Primary
Description	The admin can create, read, update, or delete records from the system's database.

Table 2.7: High Level Use Case: Manage Service Records (CRUD Operations)

Use Case (UC08)	Point Of Sale System (POS)
Actors	Clients (Users), Receptionist (Employee), System
Type	Secondary
Description	The client will pay at the end of the service and the employee will complete the transaction using the POS system. The system will update the database after generating a receipt.

Table 2.8: High Level Use Case: Point Of Sale System (POS)

2.1.2 Expanded Use Case

Use Case (UC09)	Manage Salon Operations
Scope	Managing employee records, services, and appointments
Level	User Goal
Primary Actor	Salon Manager
Stakeholders and Interests	Salon Manager, Staff, Customers
Pre-conditions	Logged in, database available
Post-conditions	Records managed
Success Guarantee	Modifications
Main Success Scenario	Manage employees, services, and appointments.

Table 2.9: Expanded Use Case: Manage Salon Operations

Use Case (UC10)	Automated Deal Generation
Scope	Generating promotional deals based on sales data and targets
Level	User Goal
Primary Actor	Salon Manager
Stakeholders and Interests	Salon Manager, Salon Owner
Pre-conditions	Logged in, sales data available
Post-conditions	Deals generated
Success Guarantee	Deals align with sales goals
Main Success Scenario	Select deal generation option and automatically view potential promotional deals

Table 2.10: Expanded Use Case: Automated Deal Generation

Use Case (UC011)	Virtual Hairstyle Try-On
Scope	Allowing users to try on hairstyles virtually
Level	User Goal
Primary Actor	Customer
Stakeholders and Interests	Customer, Salon Manager
Pre-conditions	Image following provided guidelines uploaded
Post-conditions	Hairstyle try-on results displayed
Success Guarantee	Realistic hairstyle application on image
Main Success Scenario	Select hairstyle to apply and view result.

Table 2.11: Expanded Use Case: Virtual Hairstyle Try-On

Use Case (UC012)	Virtual Hair Color Try-On
Scope	Allowing users to try on hair colors virtually
Level	User Goal
Primary Actor	Customer
Stakeholders and Interests	Customer, Salon Manager
Pre-conditions	Image following provided guidelines uploaded
Post-conditions	Hair color try-on results displayed
Success Guarantee	Realistic hair color application on image

Main Scenario	Select color to apply and view realistic results.
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Table 2.12: Expanded Use Case: Virtual Hair Color Try-On

Use Case (UC013)	Get Recommendations
Scope	Providing hairstyle recommendations based on facial features
Level	User Goal
Primary Actor	Customer
Stakeholders and Interests	Customer, Salon Manager
Pre-conditions	Image following provided guidelines uploaded
Post-conditions	Recommendations provided
Success Guarantee	Recommendations given
Main Success Scenario	Upload image, get recommended styles.

Table 2.13: Expanded Use Case: Get Recommendations

Use Case (UC014)	Appointment Booking
Scope	Booking appointments for available salon services
Level	User Goal
Primary Actor	Customer
Stakeholders and Interests	Customer, Salon Manager
Pre-conditions	Options displayed for selection
Post-conditions	Appointment confirmed
Success Guarantee	Appointment booked and stored in database
Main Success Scenario	Select service, date and time. Appointment successfully booked.

Table 2.14: Expanded Use Case: Appointment Booking

2.1.3 Use Case Diagram

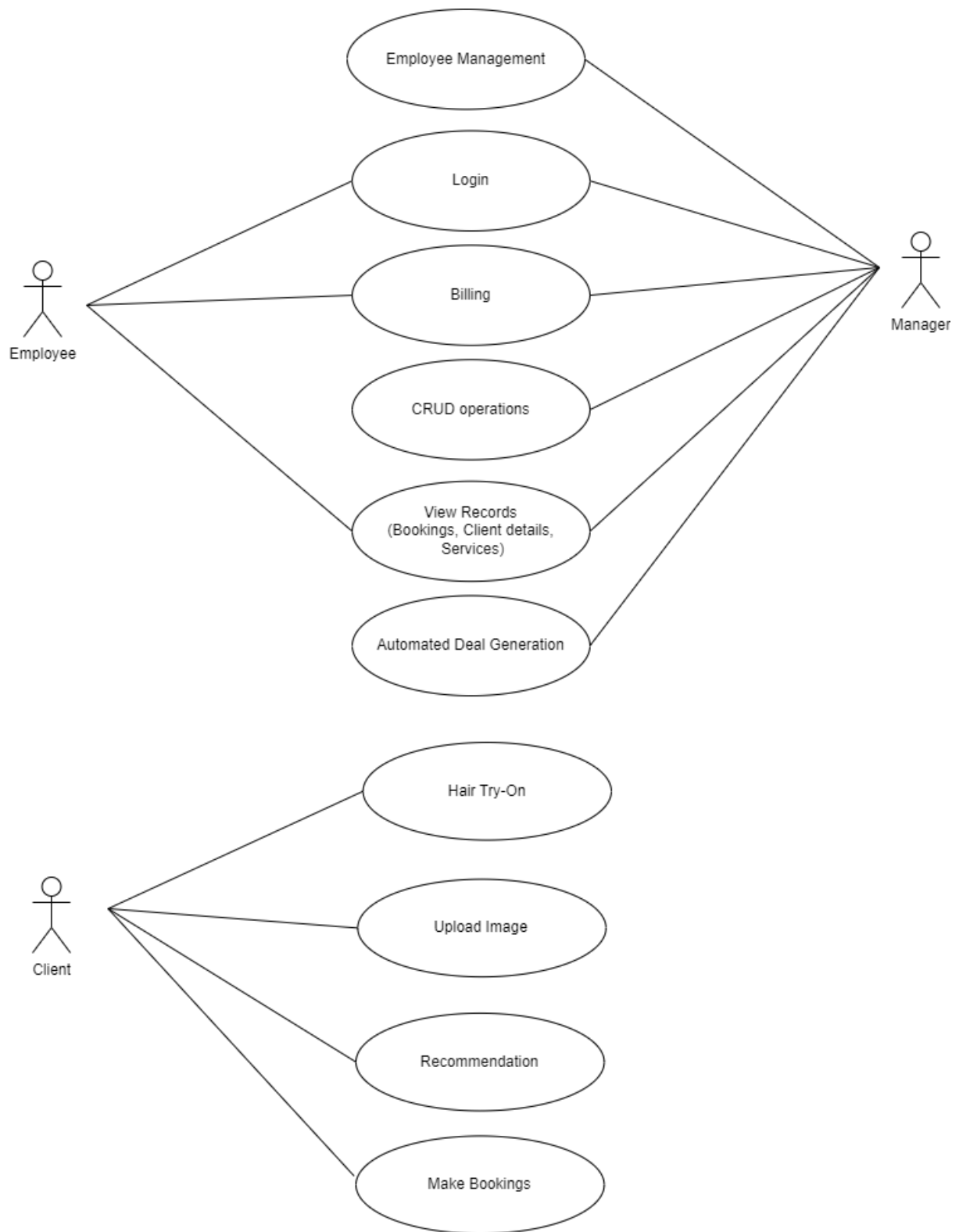


Figure 2.1: Use case Diagram.

2.1.4 Event Response Table

Event	System State	System Response
Client clicks on “Get Started”	User is on the landing page	System displays a main menu page with the three options.
Client clicks on "Book an Appointment"	Show dates and available services	System confirms the appointment and updates salon’s booking records.
Client clicks on “Get Hairstyle Recommendation”	No image uploaded	System prompts the user to upload a front-facing image without any obstructions on face and hair.
Client clicks on “Virtual Hairstyle Try-On” or “Virtual Hair Color Try-On”	No image uploaded	System prompts the user to upload a front-facing image without any obstructions on face and hair.
Client uploads a front-facing photo (for recommendation)	Photo uploaded successfully; face analysis initiated	System starts analyzing the face shape and outputs hairstyle recommendations based on the analysis.
Client uploads a front-facing photo (for virtual try-on)	Photo uploaded successfully; image processing initiated	If the selected option was "Virtual Hairstyle Try-On" or "Virtual Hair Color Try-On," the system displays available hairstyles or color options.
Client clicks on “Try-On” (for hairstyle or hair color)	Valid photo uploaded, and hairstyle or color option chosen	System applies the selected hairstyle or color to the image and displays an output.
Receptionist checks appointment status	System ready with booking data	System displays the upcoming appointments and their details.
Salon owner generates automated deals	Sales data available to the system	System triggers the automated deal generator to propose promotions or discounts to boost client engagement and sales.

Table 2.15: Event-Response Table

2.2 Functional Requirements

2.2.1 Module 1: Management Module (Admin Web App)

Following are the functional requirements for Module 1:

FR1-001: User Authentication

Field	Information
Title	User Authentication
ID	FR1-001
Description	The system provides login and sign up authentication for employees and managers before accessing the system
Priority	High
Rationale	Enables roles and permissions to the users of the system
Cross-Reference	Refer to Section 3.2.2 State Transition Diagram (Figure 3.3)

Table 2.16: FR1-001: User Authentication

FR1-002: Salon Record Management System

Field	Information
Title	Salon Record Management System
ID	FR1-002
Description	The system provides the admin with CRUD (Create, Read, Update, and Delete) functionality to manage salon operations through a centralized database. This database maintains records of customers, orders, bookings, employees, and services.
Priority	High
Rationale	Ensures efficient management of salon operations
Cross-Reference	Refer to Section 2.1.2 Use case Diagram (Figure 2.1)

Table 2.17: FR1-002 : Salon Record Management System

FR1-003: Customer Bookings

Field	Information
Title	Customer Bookings
ID	FR1-003
Description	The system provides customers with a set of available services to select from and book their appointments.
Priority	High
Rationale	Ensures timely management of customer booking
Cross-Reference	Refer to Section 2.1.2 Use case Diagram (Figure 2.1)

Table 2.18: FR1-003: Customer Bookings

FR1-004: Point of sale system

Field	Information
Title	Point of sale system
ID	FR1-004
Description	The system provides its management staff with the functionality of processing customer orders and generating invoices.
Priority	Medium
Rationale	Ensures employees to handle sales digitally.
Cross-Reference	Refer to Section 3.2.2 Management Module

Table 2.19: FR1-004: Point of sale system

FR1-005: Automated Deal Generation

Field	Information
Title	Automated Deal Generation
ID	FR1-005
Description	The system provides the admin an option to generate monthly deals and promotions based on trending and most frequently purchased services. This helps in meeting financial targets as well as maintaining customer engagement.
Priority	Medium

Rationale	Helps the admin to eliminate manual efforts required to create deals and boost sales.
Cross-Reference	Refer to Table 2.16: FR1-001: User Authentication

Table 2.20: FR1-005: Automated Deal Generation

FR1-006: Email Reminders

Field	Information
Title	Email Reminders
ID	FR1-006
Description	The system generates and send an email 24 hours prior to customers as a reminder to their upcoming booking.
Priority	Medium
Rationale	Enhances overall customer service, experience and managing missed appointments.
Cross-Reference	Refer to Table 2.16: FR1-001: User Authentication

Table 2.21: FR1-006: Email Reminders

2.2.2 Module 2: User Module (Client Web App)

Following are the functional requirements for Module 2:

FR2-001: Uploading Image

Field	Information
Title	Uploading Image
ID	FR2-001
Description	The system allows users to upload image that must be clear and without any obstruction on face and hair
Priority	High
Rationale	Ensures that the output generated creates a realistic experience for the customer.
Cross-Reference	Refer to Section 3.1 Presentation Layer

Table 2.22: FR2-001: Uploading Image

FR2-002: Hairstyle Try-On

Field	Information
Title	Hairstyle Try-On
ID	FR2-002
Description	The system allows users to virtually try-on different hairstyles and get a glimpse of their looks before their transformation.
Priority	High
Rationale	Enables customers to make bold decisions and enhance overall experience.
Cross-Reference	Refer to Section 3.1 Presentation Layer

Table 2.23: FR2-002: Hairstyle Try-On

FR2-003: Hair Color Try-On

Field	Information
Title	Hair Color Try-On
ID	FR2-003
Description	The system allows users to virtually try-on different hair colors ranging from basic to advance shades like low-lights and highlights and get a glimpse of their looks before their transformation.
Priority	High
Rationale	Enables customers to make bold decisions, preview them and increase customer satisfaction.
Cross-Reference	Refer to Section 3.1 Presentation Layer

Table 2.24: FR2-003: Hair Color Try-On

FR2-004: Recommendation Based on Facial Features

Field	Information
Title	Recommendation Based on Facial Features
ID	FR2-004
Description	The system analyzes the uploaded image to provide customers with personalised recommendation of hairstyles based on their facial features.
Priority	Medium
Rationale	Enables salons to have a customer driven sales approach and also help customers to make good decisions.
Cross-Reference	Refer to Section 3.2.2 Sequence Diagram

Table 2.25: FR2-004: Recommendation Based on Facial Features

2.3 Non-Functional Requirements**2.3.1 Reliability**

Field	Information
ID	NFR-REL-001
Title	Success Rate in Recommendation System
Description	The recommendation system's success rate should be more than 90%.
Rationale	Enables higher customer satisfaction rates.
Validation	Record the cases of lower success rates, replicate them, and conduct regular testing.

Table 2.26: NFR-REL-001: Reliability

2.3.2 Usability

Field	Information
ID	NFR-USE-001
Title	User Friendly Interface
Description	The system provides a user-friendly interface for its clients to use its virtual try-on and booking system.
Rationale	Ensures higher customer engagement and increases overall customer satisfaction rates.
Validation	Record system response time regularly and collect customer feedback to improve the system.

Table 2.27: NFR-USE-001: Usability

2.3.3 Performance

Field	Information
ID	NFR-PER-001
Title	Output Processing Time Duration
Description	The output of the virtual try-ons should be less than 10 seconds.
Rationale	Enables a smooth user experience while enhancing their virtual try-on experience.
Validation	Keep a record of system response time during peak hours and test the system with different sizes and qualities of images.

Table 2.28: NFR-PER-001: Performance

2.3.4 Security

Field	Information
ID	NFR-SEC-001
Title	Encryption of Sensitive Data
Description	The system protects user data by encryption using standards like AES-256 to build user trust and prevent security breaches.
Rationale	Ensures privacy of user details and compliance with data privacy standards.
Validation	Regular testing of system users, data transfers, system penetration testing, and suspicious activity records.

Table 2.29: NFR-SEC-001: Encryption of Sensitive Data

Field	Information
ID	NFR-SEC-002
Title	User Authentication
Description	The system provides user login functionality to ensure that only authentic users have access to the system.
Rationale	Ensures that the system is only accessible to authorized users to ensure overall data security.
Validation	Verify login flows and renew user passwords after a fixed period of time.

Table 2.30: NFR-SEC-002: User Authentication

2.3.5 Availability

Field	Information
ID	NFR-AVL-001
Title	System Availability
Description	The system must be available to its users 100% of the time for a seamless flow of events and without the loss of revenue.
Rationale	Ensures that users can access the system whenever they need it.
Validation	Record system downtime. Perform heavy load testing and maintain backup logs to ensure the latest version of records is saved on the server.

Table 2.31: NFR-AVL-001: Availability

2.3.6 Maintainability

Field	Information
ID	NFR-MAN-001
Title	System Maintenance
Description	The source code of the system must be modular and scalable to include new user requirements and upgrades with ease. Proper documentation of features must be maintained to help in troubleshooting and debugging errors.
Rationale	Ensures that the system is not disrupted due to feature enhancements and bug fixing.
Validation	Implement version control practices using machine learning operations to build and maintain a pipeline for automated updates with each new feature.

Table 2.32: NFR-MAN-001: Maintainability

2.3.7 Compatibility

Field	Information
ID	NFR-COM-001
Title	System Compatibility
Description	The system design must be compatible across multiple devices and browsers while maintaining the overall user experience.
Rationale	Ensures system design consistency across multiple screen types.
Validation	Implement dynamic sizing of components to ensure consistency in designs across multiple screens.

Table 2.33: NFR-COM-001: Compatibility

2.3.8 Scalability

Field	Information
ID	NFR-SCA-001
Title	System Scalability
Description	The system must be scalable to expand the user database and ensure a seamless experience during heavy user traffic.
Rationale	The system must ensure a balance between user growth and performance.
Validation	Implement load testing, review, and maintain the architecture of the system over time.

Table 2.34: NFR-SCA-001: Scalability

2.3.9 Accessibility

Field	Information
ID	NFR-ACC-001
Title	System Accessibility
Description	The system must be accessible to all individuals, complying with standards to cater to individuals with disabilities, enabling them to engage with the system with ease.
Rationale	Ensures better accessibility among all individuals and attracts a larger customer base.
Validation	Implement customer feedback testing on accessibility and improve it accordingly.

Table 2.35: NFR-ACC-001: Accessibility

2.4 Domain Model

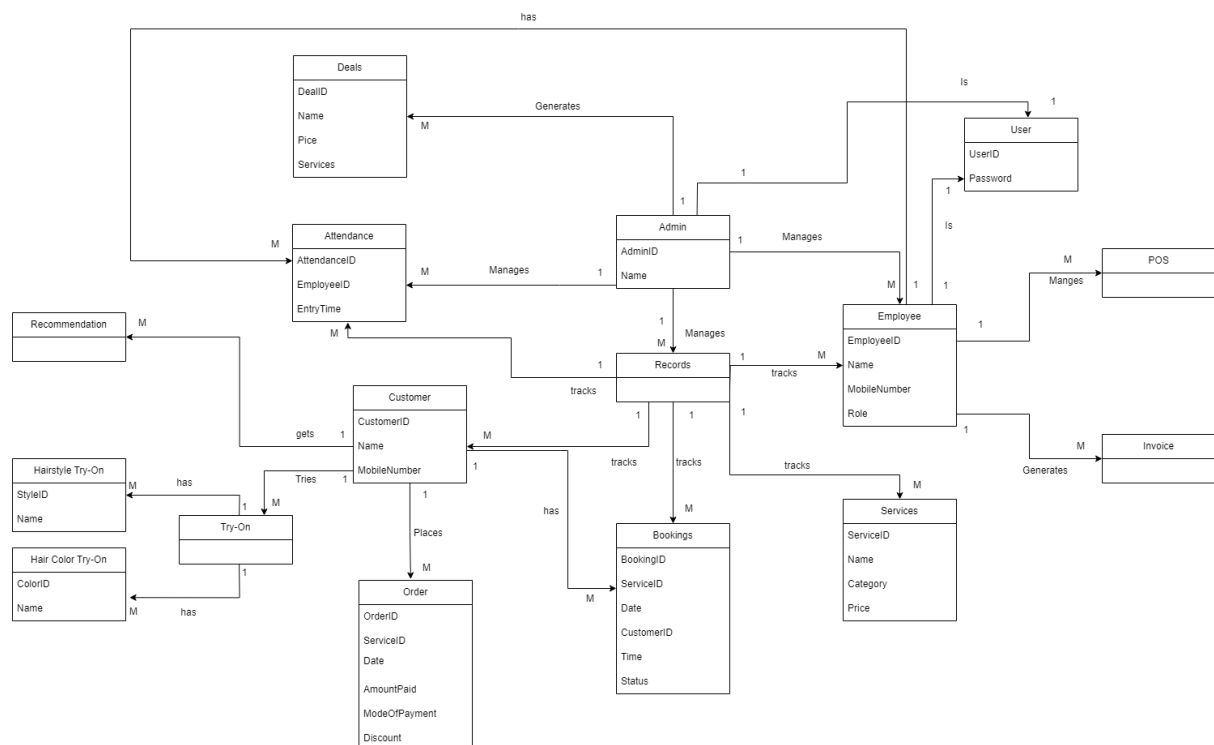


Figure 2.2: Domain Model Diagram.

Chapter 3

System Overview

3.1 Architectural Design

Multi-tiered diagram of System Architecture:

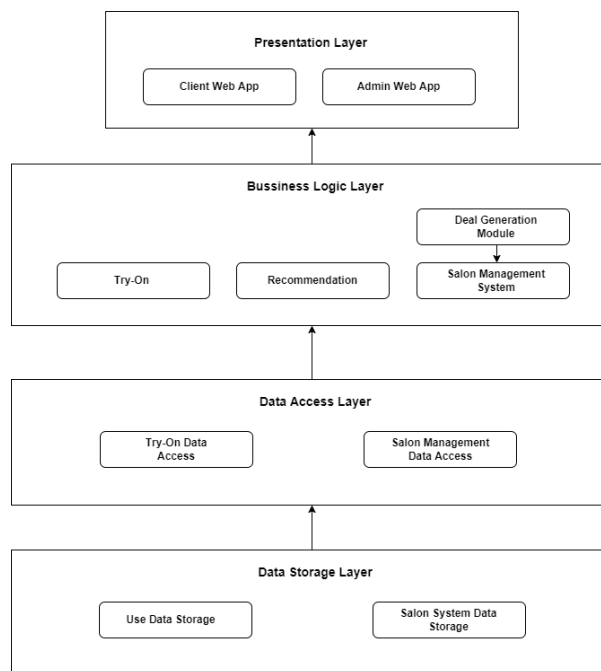


Figure 3.1: Salon360 System Architecture Diagram

Presentation Layer

- **Client Web App:** A client can upload an image, try on available hair colors and styles, get custom recommendations, and book appointments.

- **Admin Web App:** Salon managers can manage employees, services, bookings, and generate deals.

Business Logic Layer

- **Try-On:** Segment and process uploaded images for hair color and style try-on.
- **Recommendation:** Gives hairstyle recommendations based on face shape.
- **Salon Management System:** Keeps track of employees, services, bookings, and POS.
- **Deal Generation:** Automates deals based on service sale records and trends to meet financial goals.

Data Access Layer

- **Try-On Data Access:** Handles images for try-on.
- **Salon Management Data Access:** Manage salon data such as bookings, customers, employees, and services.

Data Storage Layer

- **User Data Storage:** Stores user details based on service records.
- **Salon Data Storage:** Store salon records.

3.2 Design Models

The system takes a procedural approach to design modeling, that encourages a simplified flow of events and ensures linear execution, reducing overhead time. This section describes and visualizes various design models that align with salaon360. These include:

- A Data Flow diagram to visualize the flow of the system information.
- A State Transition diagram to capture event handling and back-end processes.
- A System-Level Sequence diagram that models interactions between actors and system modules.
- An Activity diagram that pictures the flow control of the system.

3.2.1 Data Flow Diagram

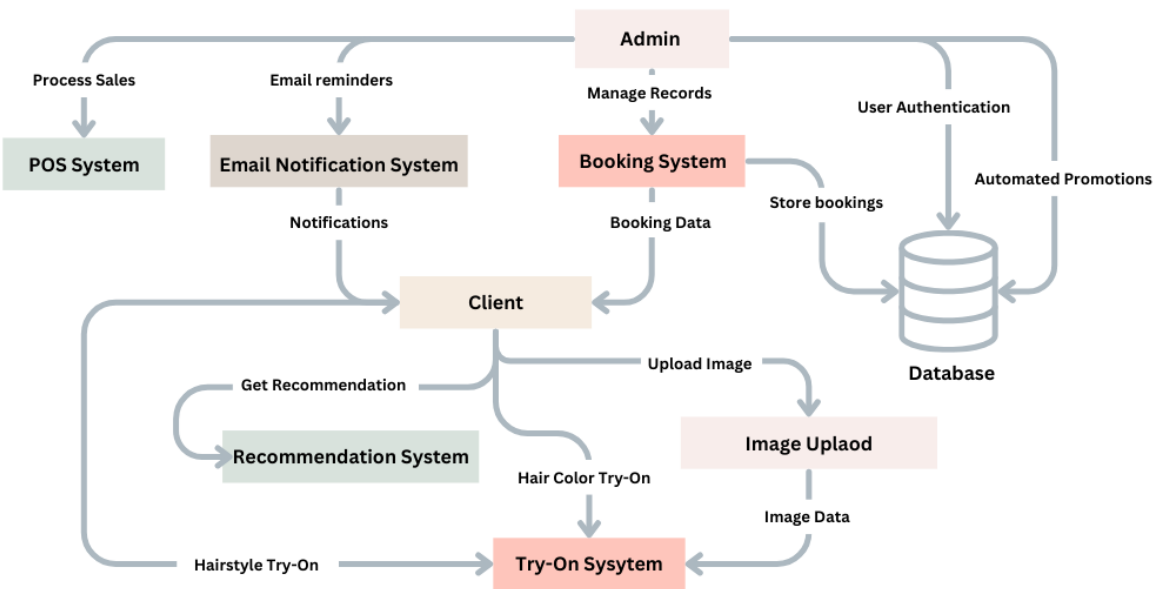


Figure 3.2: Data Flow Diagram

3.2.2 State Transition Diagram

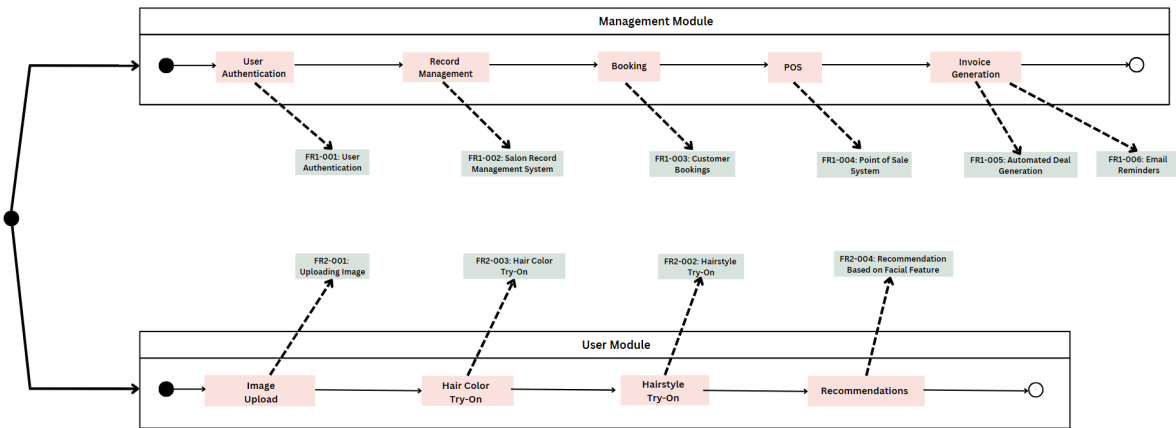


Figure 3.3: State Transition Diagram

With reference to figure 3.3 State Transition Diagram

- **Management module** starts its initial state with user authentication. After login, it allows to views records, manage bookings, process order and generate invoices and additionally allows admin to generate automated deals.
- **User module** starts its state with image upload button that leads them to two option either to get recommendations or to virtually try-on hairstyles and colors. Upon selection they can schedule appointments.

3.2.3 Sequence Diagram

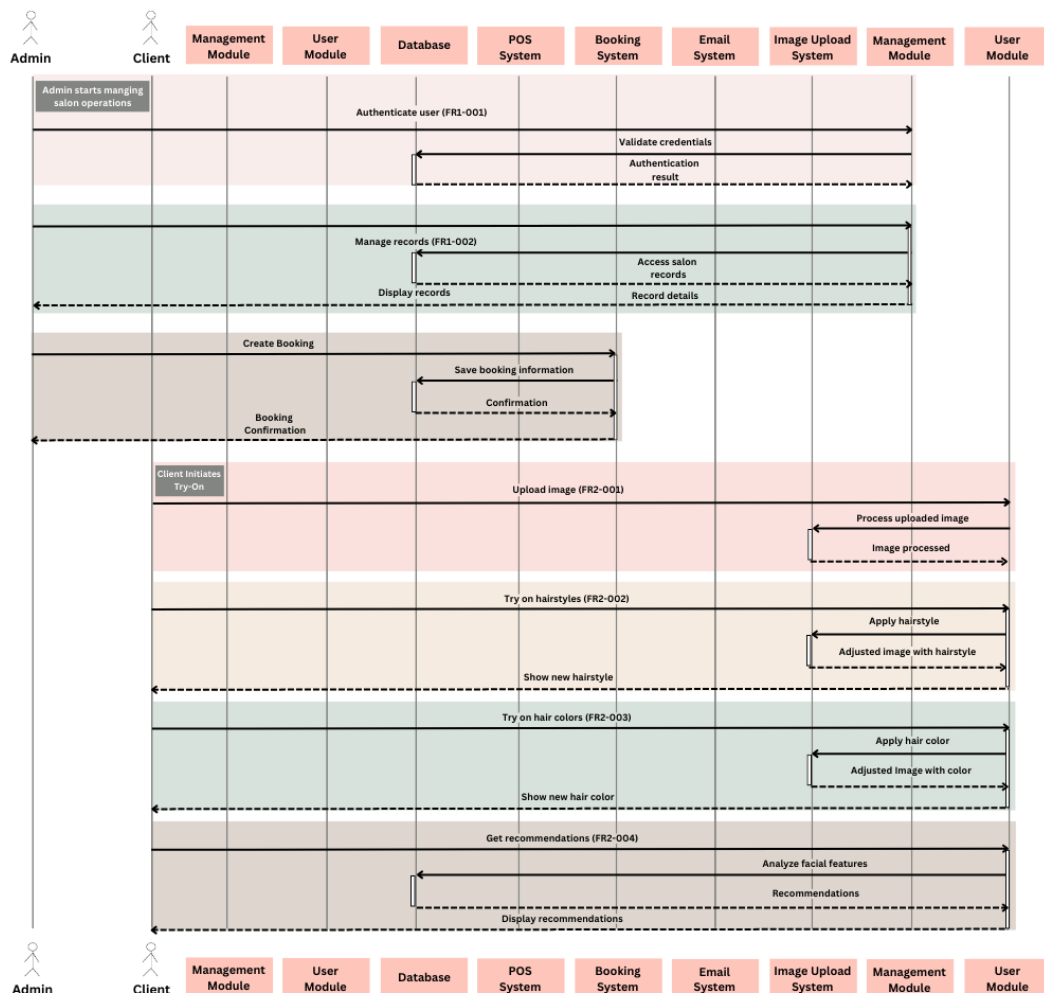


Figure 3.4: Sequence Diagram

With reference to figure 3.4 Sequence Diagram, that demonstrates the interaction between the system actors and system modules.

- **Actors:**
 - **Client** interacts with the virtual try-on engine and recommendations system.
 - **Admin/staff** interacts with the management module for seamless salon operations.
- **System Modules:**
 - **User module**
 - **Admin module**
 - **Database**
 - **Point of Sale**
 - **Booking system**
 - **Email system**
 - **Image upload system**

Processes:

- The **Admin module** is responsible for authenticating employees and allowing them to access the system and its stored records.
- **Clients** may utilize the user module to virtually try on and book hairstyles and colors.
 - The **Booking System** stores booking details in the **Database**.
 - Upon booking, the **Email system** sends out an email to the client, confirming their booking.
- The **Image upload system** manages the images uploaded by the client.
- The system allows clients to try various color and hairstyle options.
- The **User module** provides ideal hairstyles based on facial features in its recommendation system.
- The **POS system** generates an Invoice against each transaction.
 - After availing of the services, clients can make payments through the POS. Their records will be saved in the **Database**.
- The system maintains a central database that stores all the records of the system **Database**.

3.2.4 Activity Diagram

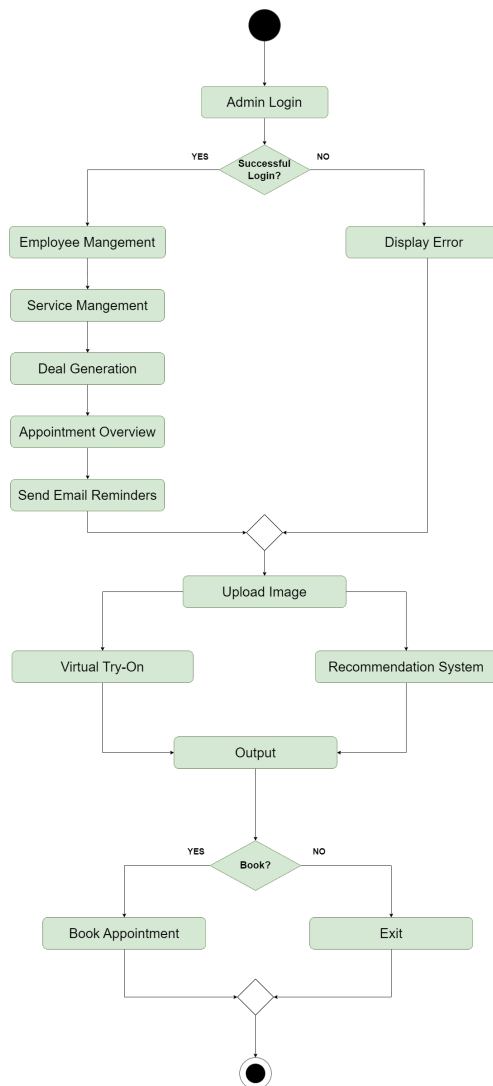


Figure 3.5: Activity Diagram

With reference to figure 3.5 Activity Diagram, the system flow is as follows:

- The system starts with a **user login authentication**. After the input there is a decision point **Successful Login?**
 - if **Yes**, then admin user can manage the system, services and generate deals. Employees can handle services, processes orders and generate invoices.

- if **No**, then system prompts user to add correct credentials (error message).
- The client-side system allows user to upload the image and choose option to either get recommendation or virtually try-on hairstyles and colors. Upon selection there is a decision point **Book?**
 - if **Yes**, then the system records the booking as an appointment with details and then exits.
 - if **No**, then the system exits.

3.3 Data Design

3.3.1 System Entities

- **Users:** Stores user ID and password for user authentication.
- **Customers:** This contains information of the customer like name, phone number and email.
- **Services:** This contains information of the services like service ID, service name, price and category.
- **Employees:** This contains information of the employees like employee ID, name and roles.
- **Attendance:** This manages the records of employee attendance like attendance ID, entry time and date.
- **Bookings:** This contains details of the appointments like booking ID, service name, booking date.
- **Order:** This contains information of the order placed like order ID, service name, order date, amount paid, mode of payment and discount.
- **Order Details:** This contains order details like order ID, service ID, quantity and price.
- **Deals:** This contains details of the deals available like deal ID, deal name and price.
- **Colors:** This contains details of the hair colors available like color ID, color name.

3.3.2 Database storage and organization

With reference to figure 2.2 Domain Model Diagram, the data of our system is organized in relational database tables (MYSQL). Each entity will be a table in the database referenced by its primary keys and linked to other tables through foreign keys. This will help to maintain the referential integrity of data and efficient querying of data.

3.3.3 Data processing and Flow of Events

With reference to figure 3.2 Data Flow Diagram,

- The system has two types of users for the management module:
 - **Admin:** The system is over all managed by the admin, the central controller who overlooks employees, services, and deal generation.
 - **Employees:** The services are handled by the employees, they process the orders placed to generate invoices.
- **Customers** select services and book appointments via the booking system, they can upload their image and get recommendation or virtually try-on different hairstyles and colors.
 - **The virtual try-on system:** is synced with the booking which allows customers to try and book their hairstyles and colors.
- **The Recommendation system:** analyzes facial features of customers through their uploaded image to give suggestions on ideal hairstyles that would suit.
- **Deal generation:** analyzes customer order details to get most frequent services and generates deals to satisfy customers and meet salon financial goals.
- **Email Reminders:** are sent to customers one day prior to their bookings

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